AI-Driven Workplace Hazard Classification

Transforming occupational safety from reactive audits to proactive, data-driven prevention using predictive machine learning to protect worker lives.

The Critical Safety Challenge

The Core Problem

In industrial manufacturing and informal labour sectors across India, unsafe working conditions often remain invisible until a severe accident or health crisis strikes. Workers face invisible threats daily—poor ventilation, extreme temperatures, inadequate protective gear—that accumulate into life-threatening situations.

The Devastating Consequence

Reactive safety measures lead to unacceptable rates of occupational hazards, fatal accidents, and chronic health issues for workers. Companies face crippling legal penalties, compensation claims, and reputational damage. The human cost is immeasurable.

Risk Categorisation

Train AI to classify workplaces into Safe, Moderate Risk, High Hazard, and Critical Violation tiers

Data Integration

Process mixed datasets combining numerical (temperature, noise) and categorical (machinery type, ventilation) variables

Automated Auditing

Enable instant hazard assessment through simple workspace metric input

Resource Allocation

Prioritise physical inspections for high-risk and critical-violation sites

Data Acquisition & Feature Engineering

Building a comprehensive dataset from occupational health sources including ILO, OSHA, and India's Ministry of Labour reports, combined with synthetic audit data to ensure robust model training.

Numerical Features

- Hours worked per week
- Ambient Temperature (°C)
- Noise Levels (dB)
- Illumination intensity

Categorical Features

- Ventilation Quality (Poor/Fair/Good)
- Machinery Type (Heavy/Light/Manual)
- Protective Gear Access (Yes/Partial/No)
- Emergency Protocol Presence

Target Variable

Four distinct hazard categories for classification:

1. Safe
2. Moderate Risk
3. High Hazard
4. Critical Violation

Preprocessing Strategy

Implement One-Hot Encoding to translate categorical data into numerical format whilst standardising numerical features. This ensures the AI can process diverse data types efficiently and accurately.

Quality Assurance

Clean missing values, remove outliers, and validate data integrity before model training to ensure reliable predictions.

Machine Learning Pipeline

01

Data Splitting

Implement 80/20 train-test split to validate model accuracy on unseen factory profiles

02

Primary Algorithm

Deploy Random Forest Classifier—ideal for handling mixed data types and capturing complex relationships between variables

03

Comparative Analysis

Benchmark against XGBoost and Support Vector Machines to identify highest accuracy model

04

Model Evaluation

Measure success using Precision and Recall metrics, emphasising Recall to prevent misclassifying Critical Violations as Safe

Key Insight: Random Forest excels at identifying complex safety patterns—e.g., "High temperatures are acceptable IF ventilation is good, but become Critical IF ventilation is poor."

Transforming Workplace Safety

AI-driven hazard classification delivers measurable benefits for workers, businesses, and regulatory bodies—shifting from reactive responses to proactive prevention.

Proactive Life Saving

Identifies deadly combinations of working conditions before they trigger industrial accidents or chronic health degradation

Optimised Inspections

Enables regulatory bodies with limited staff to direct physical inspectors only to highest-risk factories

Cost Reduction

Businesses avoid severe regulatory fines, lawsuits, and inflated workers' compensation premiums by maintaining Safe classifications

Workforce Empowerment

Provides standardised, unbiased mathematical metrics for labour unions and workers to demand better conditions using hard data

---

For Workers

Standardised, unbiased safety metrics to demand better conditions based on hard data rather than subjective complaints.

For Management

Avoid severe regulatory fines, lawsuits, and high insurance premiums by maintaining "Safe" classifications.